Problem 1 Vertical line

Read a positive integer n. Use a for loop to print a vertical line of n stars, one star per line.

1 Pseudocode

Initialize n to 0

Cout and cin for n

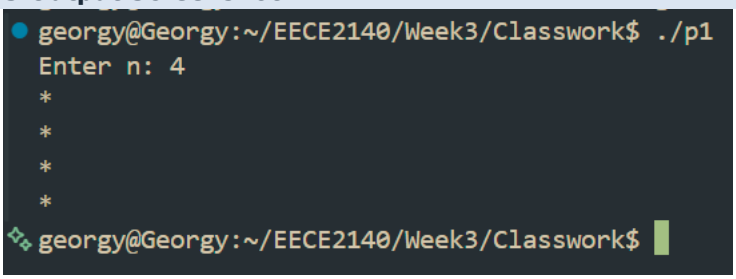
For loop i = 0 to n-1

Print * and endl

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 2 Horizontal line

Read a positive integer n. Use a for loop to print n stars on one line with no spaces between them. Compare your loop with Problem 1: what is the only line that changed?

1 Pseudocode

Initialize n to 0

Cout and cin for n

For loop i = 0 to n-1

Print * with no endl

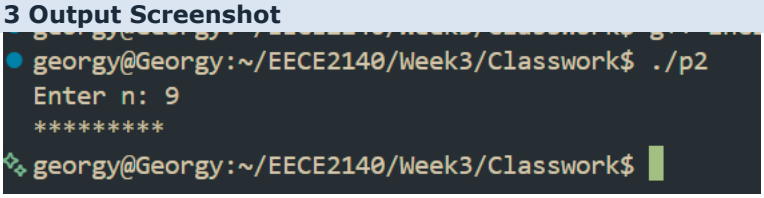
Endl after the loop

Only line that changed from Problem 1 is moving the endl out of the loop

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 3 L-shape

Read a positive integer n. Print a vertical line of n - 1 stars followed by a horizontal line of n stars, so the two lines form the letter L. Use two separate for loops.

1 Pseudocode

Initialize n to 0

Cout and cin for n

For loop i = 1 to n-1, print * and endl

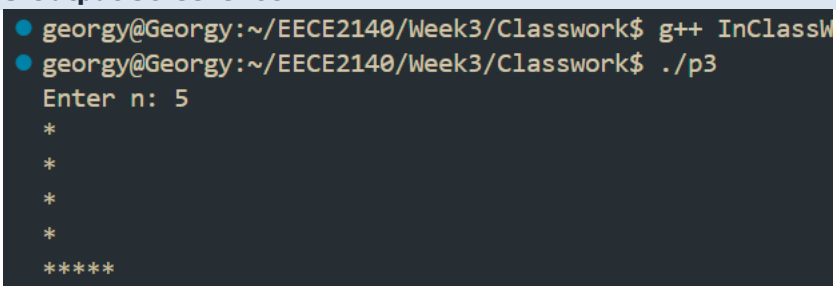
Second for loop i = 0 to n-1, print * on same line

Endl at the end

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 4 Solid rectangle

Read the number of rows and columns. Use nested for loops to print a solid rectangle of stars. The outer loop controls the rows; the inner loop controls the columns.

1 Pseudocode

Initialize row and column to 0

Cout and cin for rows and columns

Outer for loop i = 0 to row-1

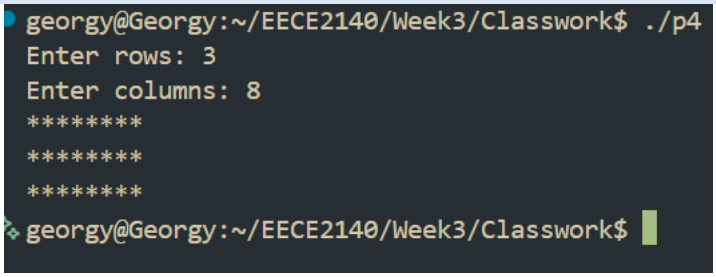
Inner for loop j = 0 to column-1, print *

Endl after inner loop done

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 5 Striped rectangle

Read the number of rows and columns. Print a rectangle in which odd-numbered rows are made of '*' and even-numbered rows are made of '-'. Use an if / else inside the outer loop with the % operator to choose the character.

1 Pseudocode

Initialize row and column to 0

Cout and cin for rows and columns

Outer for loop i = 0 to row-1

If i%2 == 0, inner for loop j = 0 to column-1, print *

Else inner for loop j = 0 to column-1, print -

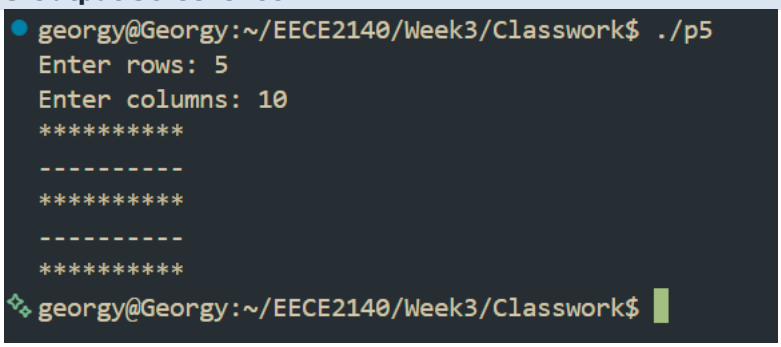
i starts at 0 so first row (odd numbered row) gets stars

Endl after inner loop

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 6 Right triangle

Read a positive integer n. Print a right triangle of n rows: row 1 has 1 star, row 2 has 2 stars, and row n has n stars. The limit of the inner loop must depend on the outer loop counter.

1 Pseudocode

Initialize n to 0

Cout and cin for n

Outer for loop i = 0 to n-1

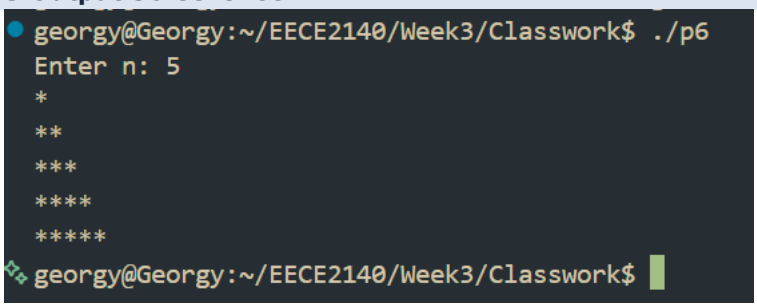
Inner for loop j = 0 to i, print *

Endl after inner loop

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 7 Inverted right triangle

Read a positive integer n. Print the triangle from Problem 6 upside down: n stars in the first row and one star fewer in each following row. Make the outer loop count down.

1 Pseudocode

Initialize n to 0

Cout and cin for n

Outer for loop starts at i = n, counts down to 1, i--

Inner for loop j = 1 to i, print *

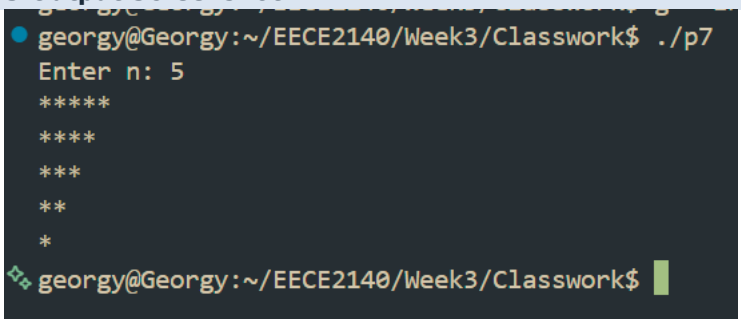
So each row prints i stars, one less every row

Endl after inner loop

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 8 Right-aligned triangle

Read a positive integer n. Print a triangle whose right edge is straight. Each row needs two inner loops: one that prints n - i spaces and one that prints i stars. Do not print spaces after the last star of a row.

1 Pseudocode

Initialize n to 0

Cout and cin for n

Outer for loop i = 0 to n-1

First inner loop starts at k = n-1, counts down while k > i, print space

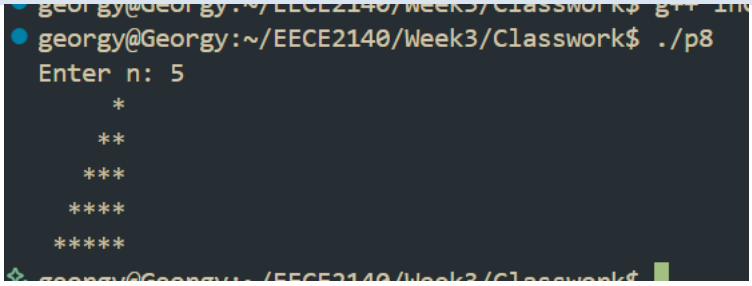
Second inner loop j = 0 to i, print *

Endl right after last star so no extra spaces

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 9 Inverted right-aligned triangle

Read a positive integer n. Print the triangle from Problem 8 upside down, so the first row has n stars and the right edge stays straight. Write a small table of row number, spaces and stars before you code.

1 Pseudocode

Initialize n to 0

Cout and cin for n

Table: row 1 = 0 spaces n stars, row 2 = 1 space n-1 stars, row n = n-1 spaces 1 star

Outer for loop i = 0 to n-1

First inner loop j = 0 to i-1, print space, so i spaces

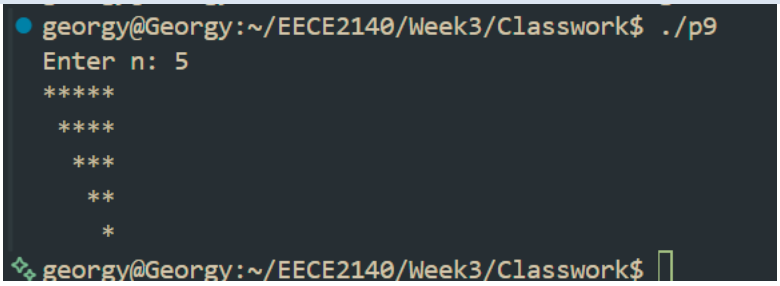
Second inner loop starts at j = n, counts down while j > i, print *, so n-i stars

Endl after each row

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 10 Number triangle

Read a positive integer n (at most 9). Print a triangle of digits where row i contains the numbers 1 through i with no spaces. Print the inner loop counter instead of a star.

1 Pseudocode

Initialize n to 0

Cout and cin for n

While n >= 10, cout re-enter n less than 10, cin n

Outer for loop i = 1 to n

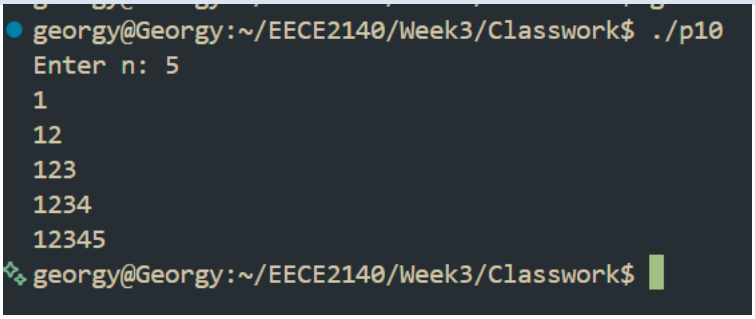
Inner for loop j = 1 to i, print j instead of *

Endl after inner loop

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 11 Floyd's triangle

Read a positive number of rows. Print Floyd's triangle: consecutive integers starting at 1, with i numbers in row i, separated by one space (no space after the last number). Use a counter variable declared before the loops.

1 Pseudocode

Initialize n to 0 and number to 1

Cout and cin for n

Outer for loop i = 0 to n-1

Inner for loop j = 0 to i, print number

If j < i print a space so no space after last number

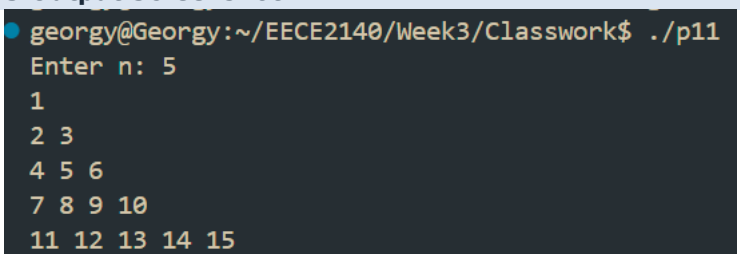
Number++ after each print

Endl after inner loop

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 12 Pyramid

Read a positive integer n. Print a centered pyramid of n rows. Row i has n - i leading spaces followed by 2 * i - 1 stars.

1 Pseudocode

Initialize n to 0

Cout and cin for n

Outer for loop i = 1 to n

First inner loop k = 0 to n-i-1, print space, so n-i spaces

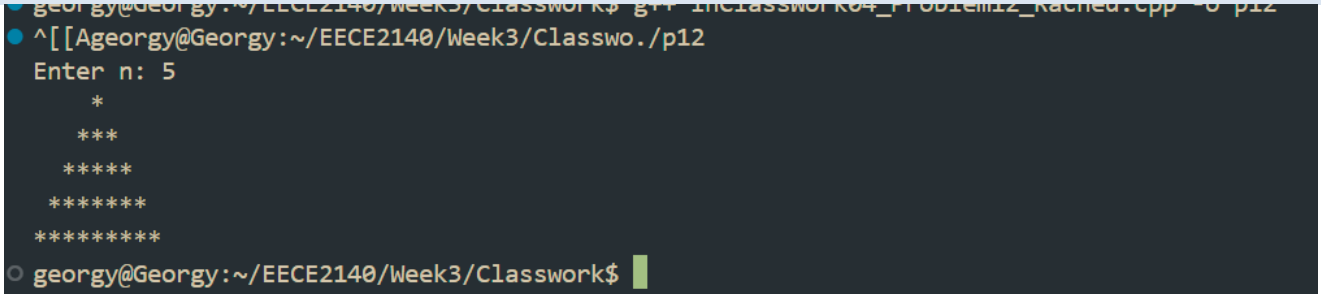
Second inner loop j = 0 to 2*i-2, print *, so 2*i-1 stars

Endl after each row

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 13 Inverted pyramid

Read a positive integer n. Print the pyramid from Problem 12 upside down, starting with the widest row of 2 * n - 1 stars.

1 Pseudocode

Initialize n to 0

Cout and cin for n

Outer for loop i = 0 to n-1

First inner loop j = 0 to i-1, print space, so i spaces

Second inner loop k = 0 to 2*(n-i)-2, print *, so 2*(n-i)-1 stars

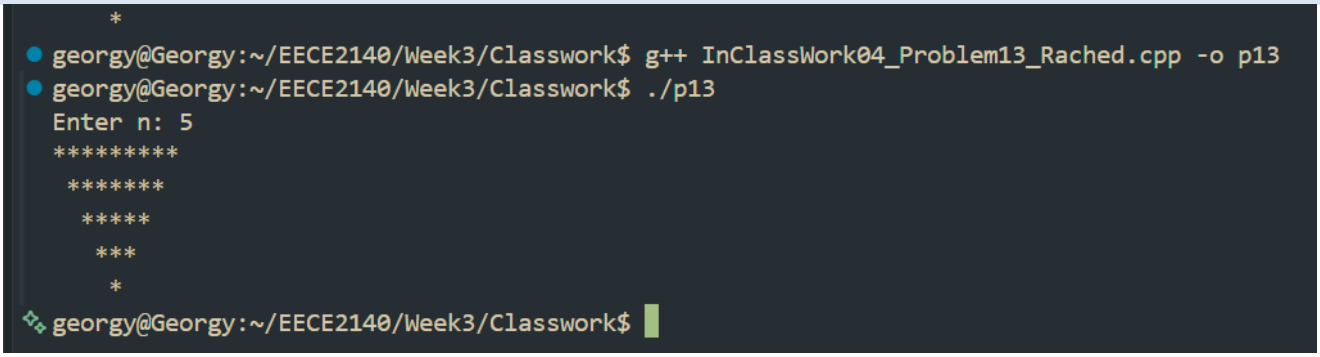
First row (i = 0) is the widest with 2*n-1 stars

Endl after each row

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 14 Diamond

Read a positive integer n. Print a diamond made of a pyramid of n rows followed by an inverted pyramid of n - 1 rows, so the widest row appears only once.

1 Pseudocode

Initialize n to 0

Cout and cin for n

Top half: pyramid from Problem 12, for loop i = 1 to n

Inner loops print n-i spaces and 2*i-1 stars, endl after each row

Bottom half: inverted pyramid from Problem 13, for loop i = 1 to n-1

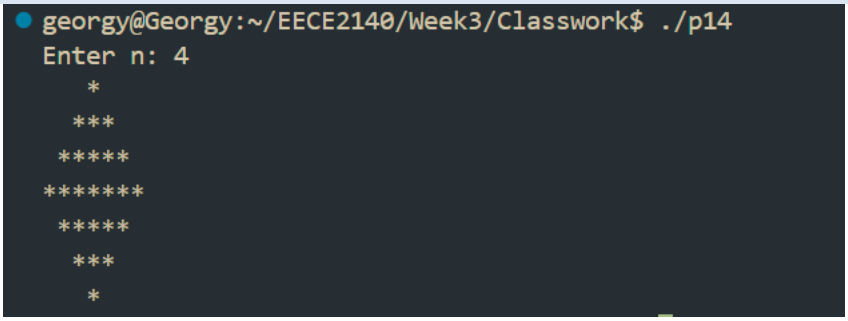
Inner loops print i spaces and 2*(n-i)-1 stars, endl after each row

Bottom starts at i = 1 (not 0) so widest row only prints once

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 15 Hourglass

Read a positive integer n. Print an hourglass: an inverted pyramid of n rows followed by a pyramid of n - 1 rows (starting at 3 stars), so the single-star row appears only once.

1 Pseudocode

Initialize n to 0

Cout and cin for n

Top half: inverted pyramid from Problem 13, for loop i = 0 to n-1

Inner loops print i spaces and 2*(n-i)-1 stars, endl after each row

Bottom half: pyramid from Problem 12, for loop i = 2 to n

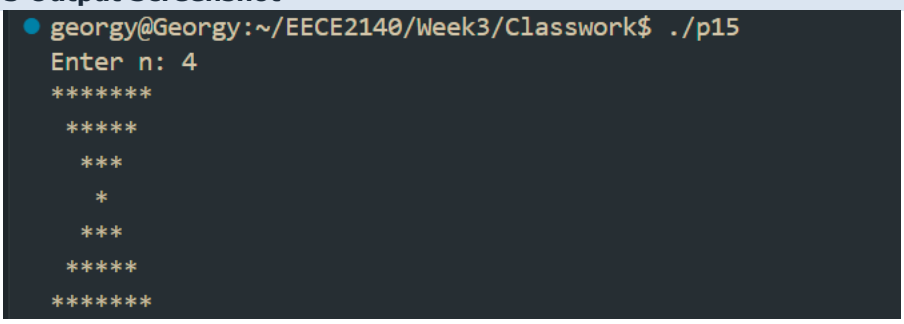
Inner loops print n-i spaces and 2*i-1 stars, endl after each row

Bottom starts at i = 2 (3 stars) so single star row only prints once

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 16 Hollow rectangle

Read the number of rows and columns. Print only the border of the rectangle. Inside the inner loop, use an if / else with || to print '*' on the first row, last row, first column or last column, and a space everywhere else.

1 Pseudocode

Initialize row and column to 0

Cout and cin for rows and columns

Outer for loop i = 0 to row-1, inner for loop j = 0 to column-1

If i == 0 || i == row-1 || j == 0 || j == column-1 print *

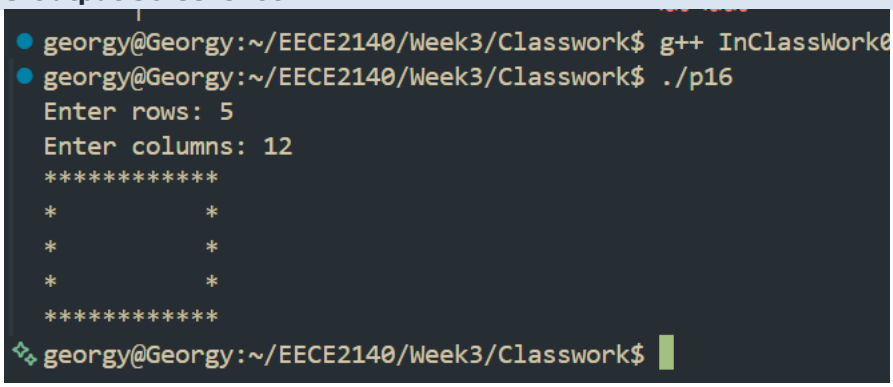
Else print space

Endl after inner loop

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 17 Hollow right triangle

Read a positive integer n. Print the outline of the right triangle from Problem 6: a star is printed only in the first column, on the diagonal (j == i), or in the last row.

1 Pseudocode

Initialize n to 0

Cout and cin for n

Outer for loop i = 0 to n-1, inner for loop j = 0 to i

If j == 0 || j == i || i == n-1 print *

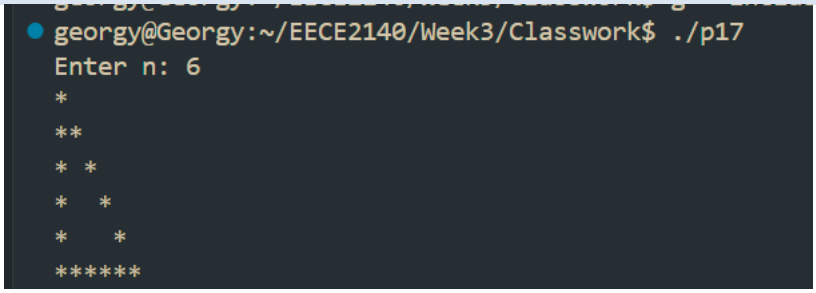
Else print space

Endl after inner loop

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 18 Hollow pyramid

Read a positive integer n. Print the outline of the pyramid from Problem 12: in each row print the leading spaces, then '*' only at the first and last position of the row, except the last row, which is completely filled.

1 Pseudocode

Initialize n to 0

Cout and cin for n

Outer for loop i = 1 to n

First inner loop k = 0 to n-i-1, print space, so n-i spaces

Second inner loop j = 0 to 2*i-2

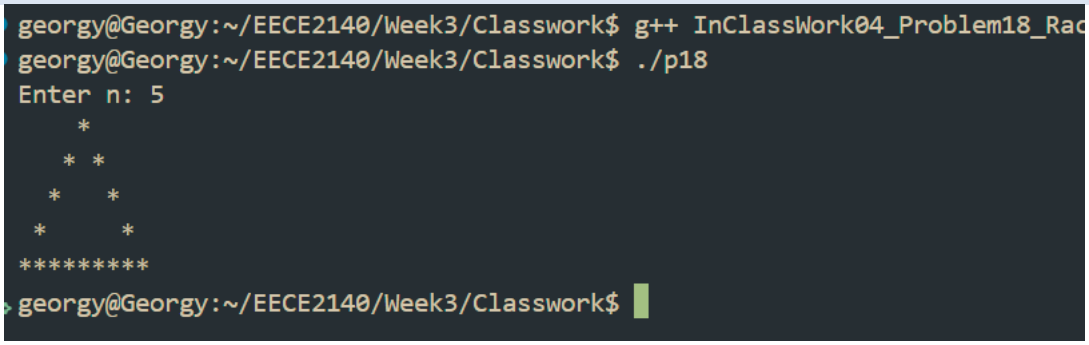
If i == n || j == 0 || j == 2*i-2 print *, else print space

Endl after each row

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 19 The letter X

Read a positive odd integer n. If n is even, print "n must be odd" and stop. Otherwise print a letter X in an n by n grid: a star belongs on a diagonal when i == j or i + j == n + 1. Do not print spaces after the last star of a row.

1 Pseudocode

Initialize n to 0

Cout and cin for n

If n%2 == 0, print "n must be odd" and return

Outer for loop i = 0 to n-1, inner for loop j = 0 to n-1

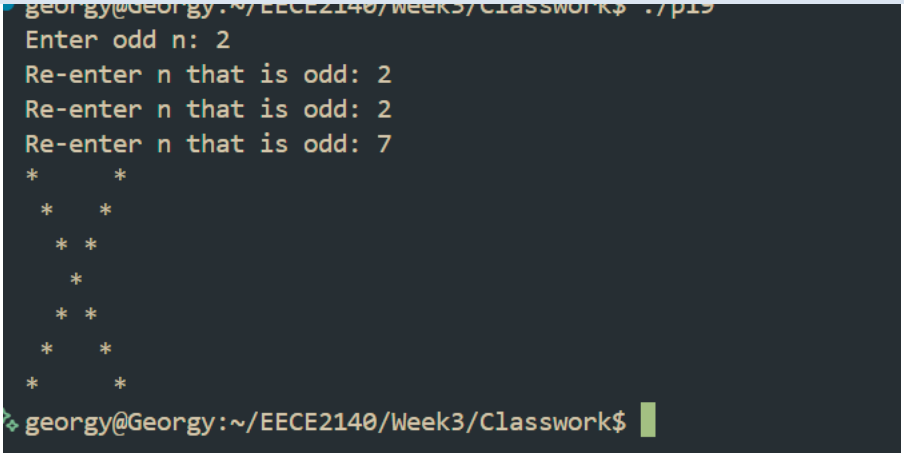
If j == i || j == n-i-1 print *, else print space

Endl after each row

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot



Problem 20 Shape menu

Combine everything: use a do-while loop to show a menu until the user enters 0. Read a choice (1 = right triangle, 2 = pyramid, 3 = diamond) and, for choices 1 to 3, a size. Use a switch statement to draw the selected shape with nested for loops. Print "Invalid choice" for any other number.

1 Pseudocode

Initialize choice and size to 0

Do while loop, runs until choice == 0

Cout menu (1 triangle, 2 pyramid, 3 diamond, 0 quit), cin choice

While choice is not 1, 2, 3 or 0, print "Invalid choice", show menu again, cin choice

Switch on choice

Case 1: cout and cin for size, right triangle loops i = 1 to size, j = 1 to i, break

Case 2: cout and cin for size, pyramid loops from Problem 12, break

Case 3: cout and cin for size, diamond loops from Problem 14, break

While choice != 0

Print "Goodbye" after loop ends

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

